

THE 697×611 RECTANGLE NEEDS EXACTLY FOURTEEN SQUARES

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ABSTRACT. Let $f(m, n)$ be the least number of integer-sided squares that tile an $m \times n$ rectangle. A 2017 answer on MathOverflow named 697×611 as the one rectangle with sides at most 760 that might violate the conjecture that scaling a rectangle never lowers f : it had a known tiling by 17 squares, while its double 1394×1222 had one by 16. We show that $f(697, 611) = 14$. The tiling is explicit; the lower bound comes from enumerating every squared rectangle with at most 13 squares through the electrical networks of Brooks, Smith, Stone and Tutte, an enumeration that reproduces all known values of f for sides up to 388.

1. THE QUESTION

Cut a rectangle with integer sides into squares with integer sides. How few squares are needed? Write $f(m, n)$ for the answer for an $m \times n$ rectangle. The values for all $m \leq n \leq 388$ are known and tabulated in the OEIS [5].

Scaling a tiling of an $m \times n$ rectangle by t tiles the $tm \times tn$ rectangle with the same number of squares, so $f(tm, tn) \leq f(m, n)$. It is natural to expect equality, and this is sometimes called the minimal squaring conjecture. In 2017, answering a MathOverflow question about the values of f [3], Ed Pegg Jr reported that among rectangles with sides at most 760 one possible counterexample remained: 697×611 , for which the best tiling known used 17 squares, while 1394×1222 could be tiled with 16 [4]. If 17 were the minimum, doubling would save a square.

Theorem 1. $f(697, 611) = 14$. *In particular 697×611 is not a counterexample to the minimal squaring conjecture.*

2. FOURTEEN SQUARES SUFFICE

Figure 1 shows the tiling. Its squares have sides

$$371, 326, 285, 240, 172, 68, 41, 37, 35, 35, 34, 34, 33, 4,$$

and their areas add up to $425\,867 = 697 \cdot 611$. With the origin at the bottom left corner, the squares sit at the positions listed in `paper/make_fig.py`, which checks cell by cell that they cover the rectangle without overlapping before it draws the figure. In Bouwkamp code, reading the squares row by row from the top, the tiling is

$$(240, 172, 285) (68, 34, 35, 35) (34) (33, 37) (371, 4) (41) (326).$$

The tiling is *compound*: it contains a smaller rectangle tiled by squares, for instance the 70×35 rectangle formed by the two squares of side 35, and it uses equal squares. Searches of catalogues of simple perfect squared rectangles, which contain neither feature, cannot find it.

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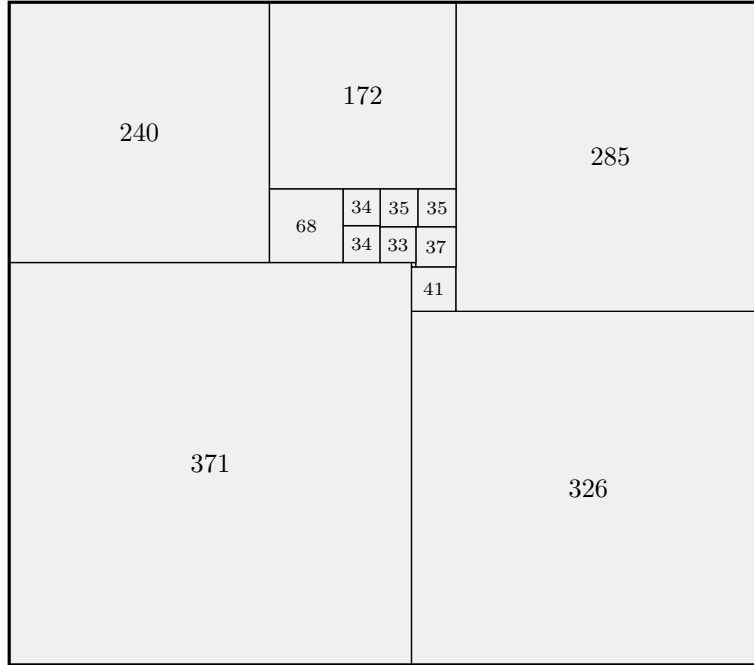


FIGURE 1. A tiling of the 697×611 rectangle by fourteen squares. Two pairs of equal squares, 35, 35 and 34, 34, sit side by side.

3. THIRTEEN SQUARES DO NOT

The lower bound uses the correspondence of Brooks, Smith, Stone and Tutte [2] between squared rectangles and electrical networks. Given a rectangle tiled by k squares, take a vertex for each maximal horizontal segment of the tiling, including the top and bottom sides, and an edge for each square, joining the segment its top lies on to the segment its bottom lies on. Add one more edge, the *pole*, from the top side to the bottom side. The result is a 2-connected plane multigraph with $k + 1$ edges. Put a unit resistor on each square's edge and a battery on the pole: the current through each edge is the side of its square, and the potential difference across the battery is the height of the rectangle. Conversely, a 2-connected plane multigraph with a chosen pole edge, in which no current vanishes, yields a squared rectangle in this way.

So every tiling of a rectangle by k squares appears among the networks with $k + 1$ edges, and to rule out a tiling it suffices to examine all of them. The 2-connected plane multigraphs with e edges are exactly one colour class of the simple quadrangulations of the sphere with e faces; the other class gives the dual networks, which produce the same tilings turned through a right angle. These quadrangulations have $e + 2$ vertices and are generated by the program *plantri* of Brinkmann and McKay [1].

Proposition 2 (computer-certified). *No network with at most 14 edges produces a tiling whose rectangle is similar to 697×611 .*

Method. The program `verify/sqrect.c` reads every quadrangulation with 5 to 16 vertices from *plantri*, takes each edge in turn as the pole, and solves for the currents. The currents are computed in floating point, scaled by the number of spanning trees and rounded, and then verified exactly in integer arithmetic: with L the Laplacian and D the number of spanning trees, the rounded potentials N must satisfy $LN = Db$. No run has failed this check. Each tiling is reduced by the greatest common divisor of its square sides and its shape recorded. Since

$697 = 17 \cdot 41$ and $611 = 13 \cdot 47$ are coprime, a tiling of 697×611 by $k \leq 13$ squares would be recorded with exactly this shape. None is: the shape first appears with 14 squares, from two networks, each giving a tiling by the fourteen squares of Figure 1. $\square$

Theorem 1 follows from the tiling and Proposition 2. Since $f(1394, 1222) \leq f(697, 611) = 14$, the doubled rectangle also needs at most fourteen squares, fewer than the sixteen previously known.

4. CHECKS

An exhaustive search is only as good as its implementation, so the same program was tested against the known values. For every rectangle with $m \leq n \leq 388$ the least number of squares it finds, taking the best over all divisors of $\gcd(m, n)$, agrees with OEIS A219158 whenever that value is at most 14, and finds no tiling when the value exceeds 14: 65 231 rectangles, no disagreement. A deliberately broken version that discards networks with parallel edges disagrees on 42 489 of the 42 862 rectangles with $f \leq 12$, so the comparison is sensitive to such errors. The tiling itself is rechecked by a separate program (`verify/verify_tiling.py`) that reads the square sides off a network, confirms the area, and finds an explicit placement by exhaustive bottom-left filling, using no network theory.

Remark. The same enumeration was run up to 16 squares (59 312 272 quadrangulations with 19 vertices). It finds Pegg’s 16-square tiling of 1394×1222 again, which serves as a positive control, and 241 further tilings of 697×611 by 16 squares. Within its range, rectangles with sides at most 1500 that can be tiled with at most 16 squares, it finds no rectangle that can be tiled with fewer squares than a rectangle it is a multiple of.

ACKNOWLEDGEMENT AND DATA

The author used an AI tool (Claude, Anthropic) to write and run the enumeration and is responsible for the content. The tiling was first posted as an answer to the MathOverflow question [3] on 29 September 2026. Programs, the OEIS comparison and the plantri 5.8 sources (Apache 2.0 licence) are in the `verify` folder accompanying this note; `verify/run.sh` reproduces Proposition 2 in under a minute on a laptop.

REFERENCES

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